

AK-RAG

Attribute Knowledge RAG — a deterministic agentic workflow for translating natural language into governed enterprise attributes.

LLM openai gpt-4o-mini

Embedding openai text-embedding-3-small (1536d)

Search local

How this pattern works

- 1 **Natural language**
User describes a population in plain English.
- 2 **LLM phrase extraction**
The LLM splits the request into individual attribute phrases — nothing else.
- 3 **Hybrid retrieval**
BM25 + kNN vector search run per phrase, fused with Reciprocal Rank Fusion.
- 4 **Decision policy**
Each phrase is classified exact

Ask for a population

diabetic patients over 65 with elevated HbA1c

high risk members with recent ER visits

patients with clinical outcomes within 12 months

patients with clinical outcomes within 12 months

Resolved DSL

Deterministic output — every entry is a real `attribute_id` from the governed catalog. Nothing here was generated freeform by the LLM.

Clinical Risk Score
clinical.risk_score

Last Clinical Visit Date
encounters.last_visit_date

```
{  
  "version": "1.0",  
  "type": "attribute_filter",  
  "filters": [  
    r
```

Each phrase is classified exact / near / ambiguous / none against fixed thresholds.

5 Clarify or resolve

Non-exact phrases ask the user to choose — the system never guesses.

6 Governed DSL

Only attribute_ids that exist in the catalog can appear in the final filter.

Catalog status

1 index(es) loaded:

attributes_demo. [Manage catalog](#)

12 MONTHS

Clinical Risk Score

clinical.risk_score

Clinical Risk Score

clinical.risk_score

Last Clinical Visit Date

encounters.last_visit_date

e.g. "diabetic patients over 65"

Evaluate

```
{
  "attribute_id":
    "clinical.risk_score",
    "business_name": "Clinical
Risk Score",
    "technical_field":
      "clinical_risk_score_num",
      "operator": null,
      "value": null,
      "source_phrase": "12 months"
},
{
  "attribute_id":
    "encounters.last_visit_date",
    "business_name": "Last
Clinical Visit Date",
    "technical_field":
      "last_visit_dt",
      "operator": null,
      "value": null,
      "source_phrase": "patients"
}
]
```

2 attribute(s) in this segment definition.